Unsupervised Human Activity Recognition using Bayesian Hidden Markov Models

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Abstract

We study unsupervised human activity recognition on the UCI smartphone HAR dataset using finite-state Gaussian hidden Markov models. Following our proposal, we compare the same Gaussian HMM under two inference strategies: classical maximum-likelihood estimation with EM and Bayesian inference with Gibbs sampling. We assume a fully unsupervised setting with no activity labels used during training. To keep Bayesian inference tractable, we use weakly informative conjugate priors: Dirichlet priors for the simplex-constrained initial and transition probabilities, and Normal-Inverse-Wishart priors for Gaussian emission parameters. A toy-data sanity check based on cached Step 2 outputs verifies that exact Forward-Backward inference and the Bayesian sampler recover the planted hidden-state structure. On the HAR benchmark, we reduce the 561 engineered features to eight principal components, treat each subject as a separate sequence, and quantify robustness by held-out log-likelihood plus clustering degradation under added Gaussian noise. The resulting comparison shows that EM slightly outperforms the Bayesian model on clustering metrics, while the Bayesian model attains slightly better held-out log-likelihood and provides posterior uncertainty over state persistence.

1 Introduction

Human activity recognition (HAR) aims to infer behaviors such as walking, sitting, and standing from wearable sensor streams. While the UCI HAR benchmark provides labels for evaluation, we treat the learning problem as fully unsupervised: labels are used only after training to measure how well the discovered latent states align with real activities.

Our project compares two inference strategies applied to the same underlying Gaussian hidden Markov model. The frequentist baseline estimates parameters with EM, while the Bayesian model places priors over the same transition and emission parameters and samples from the posterior with Gibbs updates. This distinction clarifies the core scientific question from the proposal: does Bayesian inference improve the practical behavior of the same sequential model, rather than changing the model class itself?

The project also addresses the concern raised in the proposal feedback about robustness. Here, robustness is defined concretely as the model's ability to maintain clustering quality when Gaussian noise is injected into the held-out test features, together with held-out log-likelihood as a measure of probabilistic fit. Uncertainty quantification matters in this setting because Bayesian inference yields posterior distributions over transitions and emissions, making it possible to inspect whether apparently persistent activities are estimated with confidence or with substantial variability.

2 Model Overview

Figure 1 shows the sequential model. For each subject-specific sequence, latent states $z_t \in \{1, \ldots, K\}$ evolve as a first-order Markov chain and emit reduced-dimensional sensor vectors $x_t \in \mathbb{R}^d$:

$$p(z_1) = \pi_{z_1}, \tag{1}$$

$$p(z_t \mid z_{t-1}) = A_{z_{t-1}, z_t}, \tag{2}$$

$$x_t \mid z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k). \tag{3}$$

We fix $K = 6$ hidden states to match the six activities in the benchmark and set $d = 8$ principal components after centering the original 561 features. This projection retains 78.7% of the total variance while keeping full-covariance Gaussian inference numerically stable.

For the EM baseline, we maximize the observed-data likelihood with exact E-steps from Forward-Backward and closed-form M-steps for π, A, μ_k, and Σ_k. For the Bayesian model, we use

$$\pi \sim \text{Dirichlet}(\alpha_\pi \mathbf{1}), \tag{4}$$

$$A_k \sim \text{Dirichlet}(\alpha_A \mathbf{1}), \tag{5}$$

$$(\mu_k, \Sigma_k) \sim \text{NIW}(m_0, \kappa_0, \Psi_0, \nu_0), \tag{6}$$

which are motivated by conjugacy and by the geometry of the parameter space: Dirichlet priors naturally constrain probability vectors to the simplex, while the Normal-Inverse-Wishart prior is the standard conjugate prior for multivariate Gaussian mean/covariance pairs. The Gibbs sampler alternates between sampling latent states with Forward-Filtering Backward-Sampling, sampling transition probabilities from Dirichlet posteriors, and sampling emission parameters from NIW posteriors.

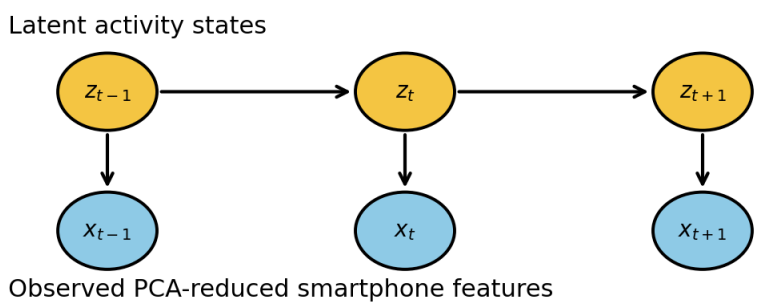

Figure 1: Graphical sketch of the Gaussian HMM used for HAR. Yellow nodes are latent activity states and blue nodes are observed PCA-reduced sensor features.

3 Experimental Setup

The UCI HAR dataset contains 10,299 sliding-window observations collected from 30 subjects performing six activities. We use the provided train/test split, center the 561 engineered features using training-set means, project onto eight principal components, and treat each subject as one sequence.

Evaluation is unsupervised: the labels are withheld during training and only used afterward for adjusted Rand index (ARI), normalized mutual information (NMI), and an aligned accuracy computed by Hungarian matching. To address the proposal feedback directly, robustness is quantified in two ways:

1. clean held-out clustering quality on the original test split, and
2. degradation in clustering quality after adding isotropic Gaussian noise to the PCA-reduced test features.

We also report held-out log-likelihood to compare probabilistic fit.

For Step 2 of the proposal, we do not regenerate synthetic results from scratch here. Instead, we use the saved toy outputs in `outputs/toy/forward_backward_results.npz` and `outputs/toy/gibbs_results.npz` which capture the exact Forward-Backward posteriors and the Gibbs samples for the three-state toy HMM.

4 Results

4.1 Step 2: Toy-data sanity check

The cached toy outputs show that the planted three-state Gaussian HMM is almost perfectly recovered. Using the exact Forward-Backward posteriors, the decoded state sequence achieves ARI 1.000, NMI 1.000, and aligned accuracy 1.000 against the true hidden states. Using the posterior mode of the Gibbs samples, the Bayesian sampler achieves ARI 0.991, NMI 0.983, and aligned accuracy 0.997. This confirms that both the exact inference and the Bayesian sampling machinery behave correctly in a controlled setting before moving to real HAR data.

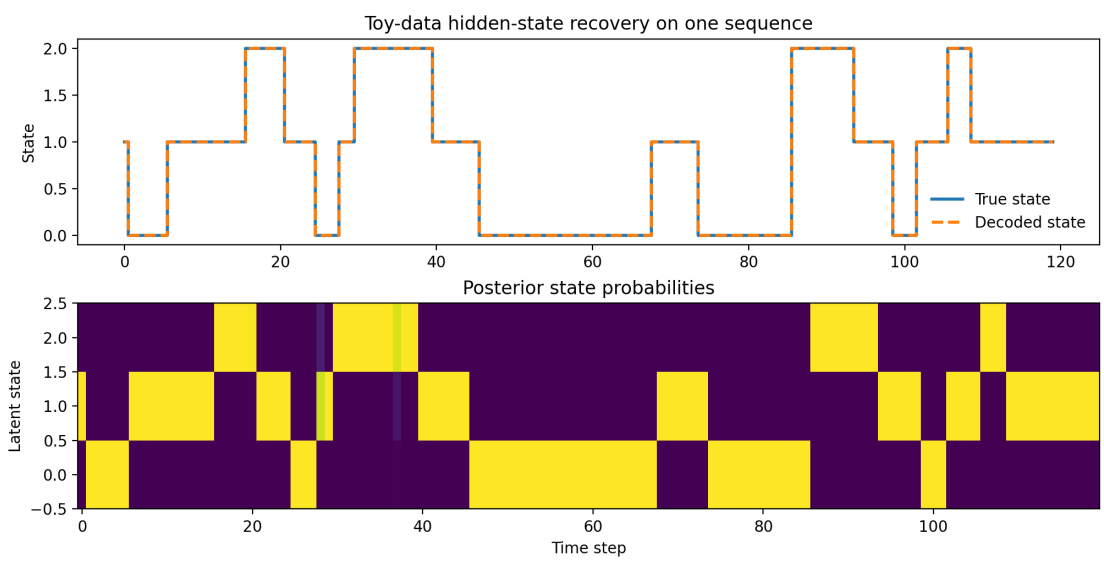

Figure 2: Step 2 toy-data sanity check. Top: true versus decoded latent states for one sequence. Bottom: posterior state probabilities from exact Forward-Backward inference.

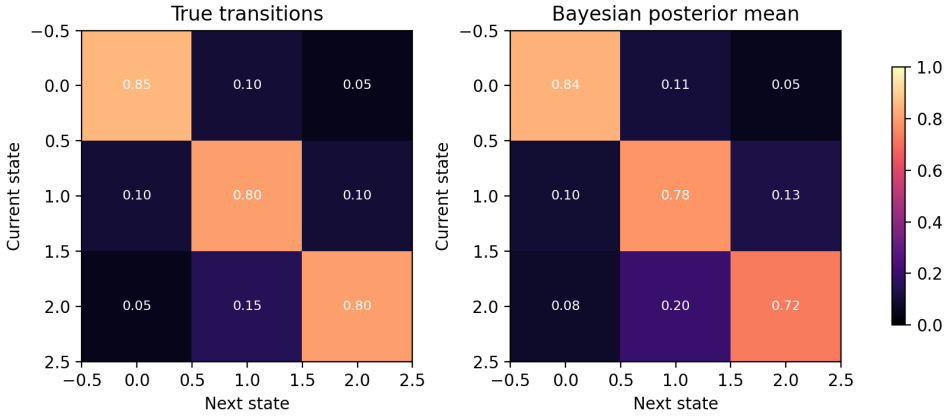

Figure 3: True toy transition matrix compared with the Bayesian posterior-mean transition matrix.

4.2 Steps 3 to 5: UCI HAR comparison and robustness

Table 1 summarizes the main comparison on held-out HAR sequences. The EM model achieved test ARI 0.301 and NMI 0.497, while the Bayesian model achieved test ARI 0.300 and NMI 0.493. The clustering scores are therefore very close, with a slight edge for EM. However, the Bayesian posterior-mean model obtained a slightly better held-out log-likelihood (-22667.7 versus -22686.1), suggesting a marginally stronger probabilistic fit even when the state assignments are not cleaner in a clustering sense.

Model	Test ARI	Test NMI	Test aligned acc.	Test log-likelihood
EM-HMM	0.301	0.497	0.448	-22686.1
Bayesian HMM	0.300	0.493	0.447	-22667.7

Table 1: Held-out clustering and likelihood comparison on UCI HAR.

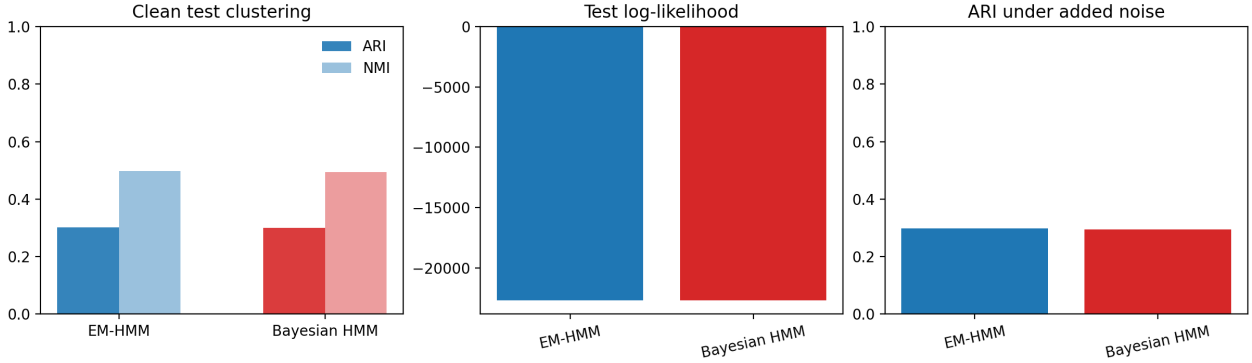

Figure 4: HAR comparison across clean clustering quality, held-out log-likelihood, and robustness under added Gaussian noise.

Our concrete robustness test adds Gaussian noise to the PCA-reduced test features. Under this perturbation, EM-HMM reaches noisy-test ARI 0.298, whereas the Bayesian HMM reaches 0.294. In other words, both methods degrade only slightly relative to the clean test set, and neither method dominates decisively on this robustness metric.

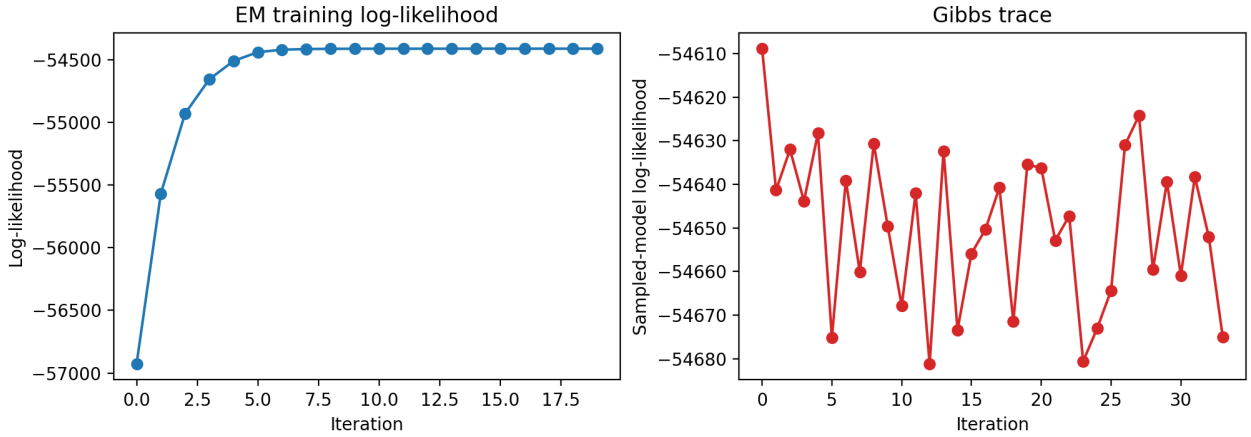

Figure 5: Optimization and sampling diagnostics. Left: EM training log-likelihood. Right: Bayesian sampled-model log-likelihood trace.

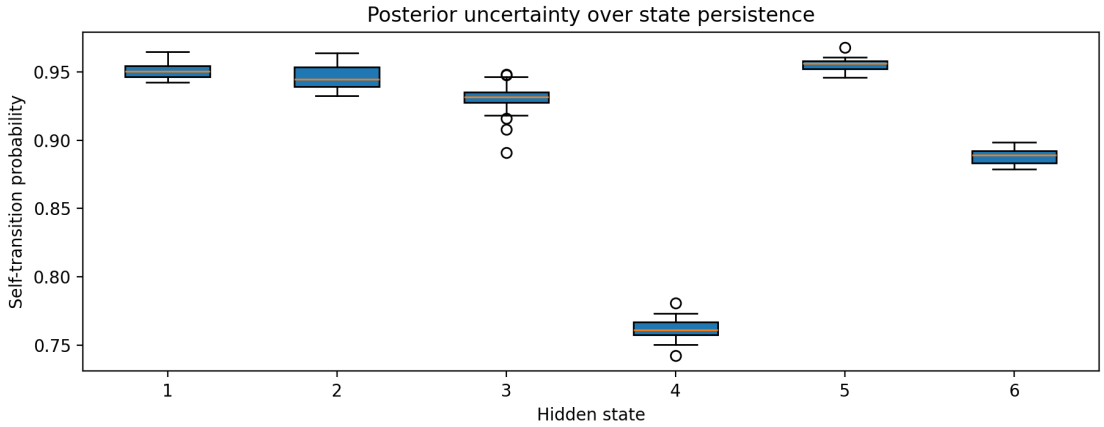

Figure 6: Posterior uncertainty over diagonal entries of the transition matrix, illustrating variation in state persistence across posterior samples.

5 Discussion

The final comparison clarifies the main narrative of the project more sharply than the proposal did. We are not comparing different model classes; rather, we are comparing EM and Bayesian inference for the same Gaussian HMM. On this dataset, Bayesian inference does not clearly improve clustering performance, but it does provide a principled posterior over transition persistence and emission parameters. That posterior is valuable when we want to understand whether long-duration states, such as sitting or laying, are estimated confidently or whether the apparent persistence is itself uncertain.

The proposal feedback also asked why uncertainty quantification should matter when the original motivation was label scarcity. The answer is that in a fully unsupervised sequential problem, uncertainty is one of the few ways to assess whether a learned latent explanation is stable. Even if the clustering metrics are similar, the Bayesian model exposes how much trust we should place in the transition dynamics.

At the same time, the results show the practical tradeoff clearly: Bayesian inference is more computationally expensive and only modestly different in predictive behavior from EM on this

benchmark. A stronger extension would be to use the raw inertial signals directly, incorporate semi-supervised labels when available, or move to a nonparametric extension such as the HDP-HMM.

6 Related Work

Rabiner (1989) remains the standard reference for HMM inference and learning, especially the Forward-Backward and Baum-Welch algorithms. Scott (2002) studied Bayesian inference for HMMs with MCMC and showed how posterior sampling gives a richer picture of uncertainty than point estimation alone. Beal (2003) developed variational Bayesian methods for graphical models, including HMMs, providing an alternative to sampling-based Bayesian inference. Teh et al. (2006) and Fox et al. (2008) extended Bayesian HMMs to nonparametric settings where the number of hidden states can be learned automatically. In the HAR domain, Anguita et al. (2013) introduced the benchmark smartphone dataset that we use here, though much of the follow-up literature has focused on supervised discriminative models rather than unsupervised sequential Bayesian modeling.

7 Conclusion

This project demonstrates that a Gaussian HMM is a reasonable unsupervised baseline for smartphone-based HAR and that both exact inference and Bayesian Gibbs sampling can be implemented cleanly for this setting. The toy-data sanity check validates the inference pipeline, and the HAR comparison shows that the frequentist and Bayesian versions of the same model perform similarly on held-out activity discovery. The main contribution of the Bayesian version is therefore interpretability and posterior uncertainty, rather than a decisive gain in clustering performance.

References

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